

C Programming in One Video

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Chapter 1 : Basics

- 1) Basic Printing, \n wali cheez.
- 2) Variables, printing variables, int, float, and +,-,*,/ of integers.
- 3) Variables naming rules.
- 4) Comments
- 5) Taking Input
- 6) Modulus Operator
- 7) Float to int, int to float
- 8) Hierarchy
- 9) Char and ASCII

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Basic program in C

```
#include<stdio.h>
```

```
int main(){
```

```
    printf("hello world");
```

```
    return 0;
```

```
}
```

VS Code

Cow has 4 legs

Cow is white

Cow gives milk

Cow is an animal

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How to move in next line?

Example :

```
printf("Hello PW");  
printf("Hello CW");
```

Output will be :

Hello PWHello CW

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Use of escape sequence '`\n`'

Output

Example :

```
printf("Hello PW");  
printf("\n");  
printf("Hello CW");
```

Output will be :

```
Hello PW  
Hello CW
```

Use of escape sequence '\n'

Predict the output : [homework]

```
main(){
    printf("nn\n\nnn\n");
    printf("nn/n/nnn/n");
}
```

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```
printf("123*2");  
printf("\n");  
✓printf("%d", 123*2);
```

↓
format
specifier

%d → for int

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Variables and their Declaration

Let us focus on int data type as of now.

1) Variables as containers :

```
✓ #include <stdio.h>
✓ int main(){
    ✓ int x;
    ✓ x = 5;
    ✓ printf("%d", x*x);
    ✓ return 0;
}
```

5
x

Output
25

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Printing Variables in C & Updation of Variables

```

✓ int x = 5;
✓ printf("%d\n", x);
✓ x = 10;
✓ printf("%d\n", x);
✓ x = x + 100; ✗
✓ printf("%d\n", x);
✓ x = x - 105;
✓ printf("%d\n", x);
✓ x = x * 2;
✓ printf("%d\n", x);
✓ return 0;
    
```

10
 8
 110
 10
 5
 x

$$x = x * 2$$

$$x = 5 * 2$$

Output

• 5
 • 10
 • 110
 • 5
 • 10

$$\begin{array}{c}
 \textcircled{10} \\
 x = x + 100; \\
 \downarrow \\
 \text{old} \\
 \text{value}
 \end{array}$$

$$\begin{aligned}
 x &= x - 105 \\
 x &= 110 - 105 = 5
 \end{aligned}$$

Arithmetic operations on int data type

```
int x = 5;
```

$$5/2 = 2$$

```
int y = 2;
```

```
printf("%d",x+y); 7
```

```
printf("%d",x-y); 3
```

```
printf("%d",x*y); 10
```

```
printf("%d",x/y); // issue
```

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Increment – Decrement operators

```
✓ int x = 5;
✓ printf("%d\n", x);
* printf("%d\n", x++);
✓ printf("%d\n", x);
```

Output

• 5
• 5
• 6

```
x--;
--x;
```

Output

• 5
• 6
• 6

x = 6

float data type

```
float x = 3.1;
```

```
int y = 5;
```

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Arithmetic operations on float data type

```
float x = 5;
```

```
float y = 2;
```

```
printf("%f",x+y);
```

```
printf("%f",x-y);
```

```
printf("%f",x*y);
```

```
printf("%f",x/y);
```

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Example : Calculating Area of a Circle

```
float radius = 5;  
float pi = 3.1415;  
float area = pi*radius*radius;  
printf("%f",area);
```

$$a = \pi r^2$$

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Example : Calculating Simple Interest

```
float p,r,t,si;  
p = 100;  
r = 10;  
t = 2;  
si = (p*r*t)/100;  
printf("%f",si);
```

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Homework : Calculate Volume of a Sphere

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Variable Naming rules

`int x;` `int raghav;`



- ✓ 1) Variables can start from an alphabet or underscore _ & \$.
- ✓ 2) Special characters except _ ~~are not allowed~~ & \$
- ✓ 3) Some particular keywords are not allowed.
- ✓ 4) Commas or blanks are not allowed.

auto
~~Auto~~ double int break extern enum unsigned while
 case sizeof for const static long continue float
 else signed do short switch char volatile default
 goto struct if union return void register typedef

Variable Naming rules – Examples

Q. Which of the following are invalid variable names and why?

BASICSALARY

_basic

basic-hra

#MEAN

group.

422

population in 2006

over time

mindovermatter

FLOAT

hELLO

queue.

team'svictory

Plot#3

2015_DDay

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Taking input // Let us take a simple example

```
int x;  
printf("Enter a number\n");  
scanf("%d",&x); // user will give 'x' a value.  
int y = x*x;  
printf("square of number that you gave is %d",y);
```

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Taking input // SUM of 2 given numbers

```
int x;  
printf("Enter first number\n");  
scanf("%d",&x); // user will give 'x' a value.  
  
int y;  
printf("Enter second number\n");  
scanf("%d",&y); // user will give 'y' a value.  
  
int sum = x+y;  
printf("sum of the numbers that you gave is %d",sum);
```


Modulus Operator

$$50 \% 2 = 1$$

$$49 \% 4 = 1$$

$$100 \% 6 = 4$$

$+, -, *, / , \%$

↓
arithmetic
operators

$a \% b$ = remainder when 'a' is divided by 'b'

↓ ↓
dividend divisor

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Modulus Operator

Miscellaneous : 1) $a \% b = a$ [if $a < b$]

$$\rightarrow 2 \% 7 = 2$$

$$2) \quad a \% (-b) = a \% b$$

$$\rightarrow 7 \% (-4) = 7 \% 4 = 3$$

$$3) \quad (-a) \% b = -[a \% b]$$

$$\rightarrow (-7) \% 4 = -[7 \% 4] = -3$$

$$4) \quad (-a) \% (-b) = -[a \% b]$$

Typecasting → 'Very Important'

✓ Ques : Take integer 'x' as input and print half of the number.

✗ Ques : Take float 'x' as input and check if it is an integer or not. → 'If Else' Good me solve karoge

Hierarchy of operators

```
int i = 2 * 3 / 4 ;
printf("%d", i);
```

Maths

$$2 * 3 / 4 = 2 * 0.75 = 1.5$$

$$2 * 3 / 4 = 6 / 4 = 1.5$$

Same

Precedence → BODMAS

Brackets > D, M > A, S

Programming

$$2 * 3 / 4 = 2 * 0 = 0$$

$$2 * 3 / 4 = 6 / 4 = 1$$

≠

Left to Right

char data type

```
char ch = 'a';
```

int , float , char
 ↓ ↓ ↓
 %d %f %c

various characters → '!', '@', '\$', '%', '^', 'a', 'b', 'A'
 '.', '>', '8'

ASCII values

```
char ch = 'a';
```



ch



```
int x = 5;
```

Each character has an ASCII value [which is an integer]

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ASCII values

yaad rehni chahiye

```
char ch = 'a';
```

'a' = 97

'b' = 98

'c' = 99

⋮

⋮

⋮

⋮

⋮

'z' = 122

'A' = 65

'B' = 66

'C' = 67

⋮

⋮

⋮

⋮

⋮

'Z' = 90

'0' = 48

'1' = 49

⋮

⋮

⋮

⋮

'9' = 57

MCQ Time !

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MCQ 1

Which of the following is NOT a character constant

- (1) 'Thank You'
- (2) 'Enter values of P, N, R'
- (3) '23.56E-03'
- (4) ☒ All the above

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MCQ 2

In $b = 6.6 / a + 2 * n$; which operation will be performed first?

☒ (1) $6.6 / a$

(2) $a + 2$

(3) $2 * n$

(4) Depends upon compiler

MCQ 3

Which of the following statements is false 'very imp'

- ✓ (1) Each new C **instruction** has to be written on a separate line **FALSE**
- (2) Usually all C statements are entered in small case letters **TRUE**
- (3) Blank spaces may be inserted between two words in a C statement **TRUE**
- (4) Blank spaces cannot be inserted within a variable name **TRUE**

MCQ 4 (Homework)

If `a` is an integer variable, `a = 5 / 2 ;` will return a value

(1) 2.5

(2) 3

✓ (3) 2

(4) 0

```
int a = 5/2;  
printf("%-d", a);
```

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MCQ 5 (Homework)

The expression, $a = 7 / 22 * (3.14 + 2) * 3 / 5 ;$ evaluates to

(1) 8.28

(2) 6.28

(3) 3.14

(4) 0

float a;

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$$a \% (-b) = a \% b$$

$$a \% b = a \quad [a < b]$$

MCQ 6

The expression $x = 4 + 2 \% - 8$ evaluates to

(1) -6

☒ (2) 6

(3) 4

(4) None of the above

$$x = 4 + 2 \% - 8$$

$$\Rightarrow 4 + 2 \% 8$$

$$\Rightarrow 4 + 2$$

$$\Rightarrow 6$$

MCQ 7

What will be the value of d if d is a float after the operation
 $d = 2 / 7$?

☒ (1) 0

(2) 0.2857

(3) Cannot be determined

(4) None of the above

float d = 2/7 ;

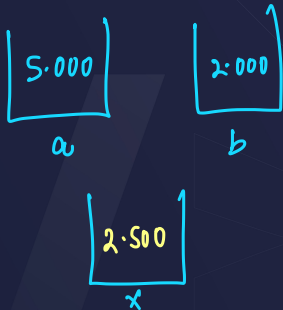
d = 2.0/7 ;

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```
float a = 5;
float b = 2;
float x = a/b;
printf("%f", x);
```

v/s

```
float x = 5/2;
printf("%f", x);
```



float / float $\rightarrow$ float
 int / int $\rightarrow$ int
 float / int $\rightarrow$ float
 int / float $\rightarrow$ float



Chapter 2 : Conditionals

- 1) if, if - else
- 2) nested
- 3) Else if ladder
- 4) Ternary
- 5) switch

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Control Statements

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```
int x;  
scanf("%d", &x);
```

IF – ELSE

↓

if statement

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```

✓int x;
✓printf("Enter a number : ");
✓scanf("%d",&x);
✓// if statement
✓if(x>0) printf("Positive Number");
✓if(x<0) printf("Negetive Number");
✓return 0;

```

8

x

Enter a number : 8

Positive Number

```

✓int x;
✓printf("Enter a number : ");
✓scanf("%d",&x);
✓// if statement
✓if(x>0) printf("Positive Number");
✓if(x<0) printf("Negetive Number");
  return 0;

```

-4
 x

Enter a number : -4
 Negetive Number

Ques : Take positive integer input and tell if it is even or odd



int x;

scanf("%d", &x);

if (even) printf("Even No");

if (odd) printf("Odd Number");

if ($x \% 2 == 0$) → even

if (condition) work ;
 ↓
 true, false

Relational Operators

$>$, $<$, $>=$, $<=$, $==$
 $!=$

Comparisons
 Equality operator

! $\rightarrow$ not

$= \rightarrow$ assignment operator

$x = x + 10;$

LHS = RHS

, LHS == RHS

$\downarrow$
 true, false

HW : Take positive integer input and tell if it is divisible by 5 or not.

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Ques : Take integer input and print the absolute value of that integer
mod

$$x = 4$$

$$|x| = |4| = 4$$

$$x = -7$$

$$|x| = |-7| = 7$$

$$\text{if } (x \geq 0 \rightarrow |x| = x)$$

$$\text{if } (x \geq 0)$$

$$\text{if } (x < 0)$$

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Typecasting → 'Very Important'

- ✓ Ques : Take integer 'x' as input and print half of the number.
- ✓ Ques : Take float 'x' as input and check if it is an integer or not. → 'If Else'

```

✓ float f;
✓ printf("Enter a real number : ");
✓ scanf("%f",&f);
✓ int x = (int)f;
✓ float a = (float)x;
✓ if(f-a == 0) False
    printf("Number is Integer");
✓ if(f-a > 0) True
    printf("Number is not Integer");
✓ return 0;
    
```

Enter a real number : 3.4

Number is not integer

3.4
f

3
x

3.00
a

Ques : If cost price and selling price of an item is input through the keyboard, write a program to determine whether the seller has made **profit** or incurred **loss** or **no profit no loss**. Also determine how much profit he made or loss he incurred.

```
int sp;    if (sp > cp)
int cp;    Profit
           sp - cp
```

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Else statement

- ↓
- used alongside/after 'if' statement
- if your 'if' statement fails, then else will run.

```
int x;  
printf("Enter a number : ");  
scanf("%d",&x);  
if(x%2==0) printf("Even Number");  
if(x%2!=0) printf("Odd Number");  
return 0;
```

↓
2 times check

```
int x;  
printf("Enter a number : ");  
scanf("%d",&x);  
if(x%2==0) printf("Even Number");  
else printf("Odd Number");  
return 0;
```

HW : Given the length and breadth of a rectangle, write a program to find whether the area of the rectangle is greater than its perimeter.

Multiple Conditions Using `&&` and `||`

and *or*

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Ques : Take positive integer input and tell if it is a three digit number or not.

```
if (x > 99 && x < 1000)  
    print (—)
```

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Ques : Take positive integer input and tell if it is divisible by 5 and 3.

$\downarrow$ $\swarrow$
 $\text{if}(x \% 5 == 0)$ $\text{if}(x \% 3 == 0)$
 $\swarrow$ $\swarrow$

$\text{if}(x \% 5 == 0 \ \&\& \ x \% 3 == 0)$

3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33...

5, 10, 15, 20, 25, 30, 35 . . .

Ques : Take positive integer input and tell if it is divisible by 5 or 3.

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Ques : Take 3 numbers input and tell if they can be the sides of a triangle.

$a, b, c \rightarrow$ sides of Δ

$$a + b > c$$

$$b + c > a$$

$$c + a > b$$

if ($a + b > c$ && $b + c > a$ && $c + a > b$)

Difference b/w && and || operators

if (condition 1 && condition 2)

true	true	→ true
true	false	→ false
false	true	→ false
false	false	→ false



if c1 is false, then compiler doesn't even check for the second condition

if (condition 1 || condition 2)

true	true	→ true
true	false	→ true
false	true	→ true
false	false	→ false



if c1 is true, then compiler doesn't even check for the c2.

* **Ques** : Take 3 positive integers input and print the greatest of them. *(distinct)*

a, b, c

$\text{if}(a > b \ \&\& \ a > c) \text{ ---}$
 $\text{if}(b > c \ \&\& \ b > a) \text{ ---}$
 $\text{if}(c > a \ \&\& \ c > b) \text{ ---}$

only works for distinct no.s

~~Ques~~ : Take 3 positive integers input and print the least of them. (*Distinct*)

Ques : Any year is input through the keyboard.
Write a program to determine whether the year is a leap year or not.

366 days

2000 ✓, 2023 ✗, 1900 ✗, 1600 ✓, 2016 ✓, 2020 ✓

1800 ✗, 1900 ✗, 1700 ✗
2000 ✓, 1600 ✓

- Leap year occurs after every 4 years (but)
- Any year divisible by 100 is not a leap year (but)
- Any year divisible by 400 is 100% a leap year

HW : Take positive integer input and tell if it is divisible by 5 or 3 but not divisible by 15.

Nested If – Else

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Ques : Take positive integer input and tell if it is divisible by 5 and 3.

```
if(x%5==0){  
    if(x%3==0) printf("Divisible by 5 and 3");  
    else printf("Not divisible by 5 and 3");  
}  
else printf("Not divisible by 5 and 3");
```

$x = 5 \rightarrow$ Not divisible by 5 & 3

$x = 3 \rightarrow$ Not divisible by 5 and 3

$x = 15 \rightarrow$ Divisible by 5 & 3

Ques : Take positive integer input and tell if it is divisible by 5 or 3 but not divisible by 15.

```
if(x%15!=0){
    if(x%3!=0){
        if(x%5==0) printf("Condition Valid");
        else printf("Condition Invalid");
    }
    else // x%3 = 0
        printf("Condition Valid");
}
else // x%15 = 0
    printf("Condition Invalid");
return 0;
```

$x = 9$: Condition Valid

$x = 20$: Condition Valid

$x = 45$: Condition Invalid

$x = 7$: Condition Invalid

***Ques** : Take 3 positive integers input and print the greatest of them. *without using (multiple conditions)*

a, b, c

if (a > b) {

if (a > c) → a is greatest

else

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HW : If the ages of Ram, Shyam and Ajay are input through the keyboard, write a program to determine the youngest of the three. *(Using nested if else)*

Else If

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Ques : Take input percentage of a student and print the Grade according to marks:

Method-1 (Multiple Condition)

- 1) 81-100 Very Good
- 2) 61-80 Good
- 3) 41-60 Average
- 4) ≤ 40 Fail

$x = 90$ Very Good

$x = 45$ Average

```
if(x>=81) printf("Very Good");  
if(x>=61 && x<=80) printf("Good");  
if(x>=41 && x<=60) printf("Average");  
if(x<=40) printf("Fail");
```

Ques : Take input percentage of a student and print the Grade according to marks:

Method - 2 Else if

- 1) 81-100 Very Good
- 2) 61-80 Good
- 3) 41-60 Average
- 4) ≤ 40 Fail

$x = 90$ Very Good

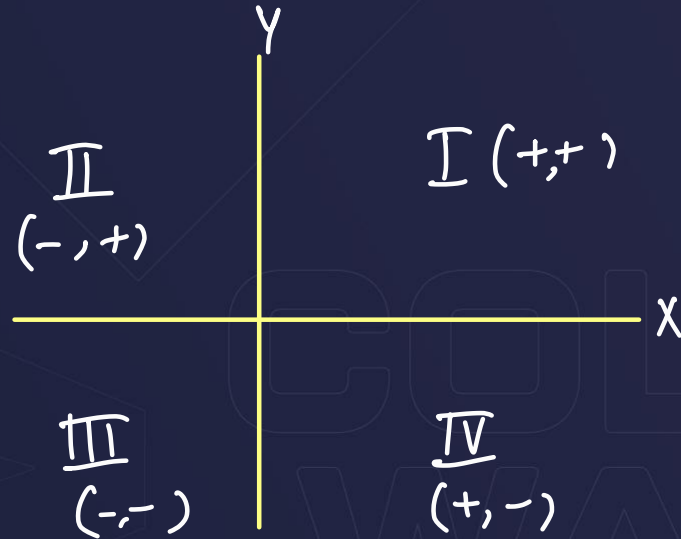
$x = 70$ Good

$x = 50$ Average

$x = 20$ Fail

```
if(x>=81) printf("Very Good");
else if(x>=61) printf("Good");
else if(x>=41) printf("Average");
else printf("Fail");
```


HW : Given a point (x, y) , write a program to find out if it lies in the 1st Quadrant, 2nd Quadrant, 3rd Quadrant, 4th Quadrant, on the x-axis, y-axis or at the origin, viz. $(0, 0)$. (Else-if)



HW : Given the coordinates (a, b) of a center of a circle and its radius, write a program which will determine whether a user entered point lies inside the circle, on the circle or outside the circle.

a, b, r
↓
given

x, y → user input

$$x^2 + y^2 = r^2$$

$$(x-a)^2 + (y-b)^2 = r^2 \rightarrow \text{on the circle}$$

$$(x-a)^2 + (y-b)^2 > r^2 \rightarrow \text{outside}$$

$$(x-a)^2 + (y-b)^2 < r^2 \rightarrow \text{inside}$$

* **Ques** : Take 3 positive integers input and print the greatest of them.

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↗ Cool Way to write if-else code

Ternary Operator

expression 1 ? expression 2 : expression 3 ;

Condition

true

false

Q₁ WAP to check if no is even or odd. using ternary.

condition ? true : false

H.W. WAP to check if input year is leap year.

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↗ Replacement for if - else
'Useless'

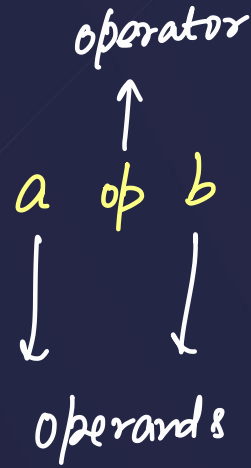
Switch Statement

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Syntax :

→ char

```
switch ( integer expression ) {
    case constant 1 :
        do this ;
    case constant 2 :
        do this ;
    case constant 3 :
        do this ;
    default :
        do this ;
}
```



$5 * 3$

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H.W: WAP to input day no. $\rightarrow$ 1, 2, 3, 4, 5, 6, 7

$\swarrow$ $\downarrow$ $\downarrow$
 mon Tue Sun

Ques : Write a program to create a calculator that performs basic arithmetic operations (add, subtract, multiply and divide) using switch case and functions. The calculator should input two numbers and an operator from user. *(done)*

Chapter - 3 : 'Loops'

- 1) For Loop
- 2) While Loop
- 3) Do-while loop

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For Loop

initialisation
condition
increment
`for(int i = 1; i < 10; i++) {`

i = i + 1

`// code`

`}`

Hello World → 10 baar

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Ques : Print hello world 'n' times. Take 'n' as input from user

```
✓int n;
✓printf("Enter n : ");
✓scanf("%d",&n);
✓for(int i=1;i<=3n;i++){
    printf("Hello\n");
}
```

4
3
2
1
i

3
n

Steps :

- 1) Check → false, finish
- 2) Work
- 3) Increment
- 4) Repeat

Output

- Enter n : 3
- Hello
- Hello
- Hello
-

How for loop works : the various parameters.

initialization ; condition ; increment
 ↓
 {x baar
 └──────────┘
 ↓
 baar baar

Ques : Print numbers from 1 to 100

1 to 10

10 times hello world

↳ loop ka variable $\rightarrow$ i

Print the value of 'i' & Then obviously 'i' is going to
increment

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Ques : Print all the even numbers from 1 to 100

→ Output

2 4 6 8 10 . . . 98 100

#Hint : If Else

this loop is running 100 times
↑

```
for(int i=1;i<=100;i++){
    if(i%2==0) printf("%d ",i);
}
```

Ques : Print all the even numbers from 1 to 100

Method-2 : By changing the increment

$i = 1, 2, 3, 4, 5, 6, \dots, 100 \rightarrow i++ / i = i + 1$

↓

$i = 2, 4, 6, 8, 10, \dots, 98, 100$

So baar loop chlega

```
for(int i=2; i<=100; i=i+2){  
    printf("%d ", i);  
}
```


HW : Print all the odd numbers from 1 to 100

↓

M-1, M-2

↓

if else

↪ increment change

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Ques : Print the table of 19.

Output : 19, 38, 57, 76, 95, . . . 171, 190

M-I : divisible by 19 [1-190]
 ↳ 190 times

M-2 → 10 times only

```
for(int i=19; i<=190; i=i+19){
    printf("%d ", i);
}
```

209
 i = 19 38 57 76 95 114 123 142 161 180

19
 38
 57
 76
 95
 114
 133
 152
 171
 190

HW : Print the table of 'n'. Here 'n' is a integer which user will input.

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Ques : Display this AP – 1,3,5,7,9.. upto 'n' terms.

```

for(int i=1;  $i \leq 2n-1$ ; i=i+2) {
    |      printf("%d\n", i);
    |
    3

```

technically I was
printing the value
of 'i'

$$a_n = a_1 + (n-1)d$$

$$a_n = 1 + (n-1) \cdot 2$$

$$a_n = 1 + 2n - 2$$

$$a_n = 2n - 1$$

Ques : Display this AP – 1,3,5,7,9.. upto 'n' terms. Without the n^{th} term formula

I will only use 'i' as the number of rounds for the loop

HW : Display this AP – 4,7,10,13,16.. upto 'n' terms.

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Ques : Display this GP – 1,2,4,8,16,32,.. upto 'n' terms.

→ I will run the ^{'loop'} code 'n' times

HW : Display this GP – 3,12,48,.. upto 'n' terms.

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Break;

↓

loop ko finish karna before reaching the condition

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Q, Find the highest factor of a given number 'n'. [except 'n']

→ For ex: $n = 24$

→ 1, 2, 3, 4, 6, 8, 12, 24

basic approach

```
→
    int hf;
    for(i=1; i<=n-1; i++){
        if(n%i==0) hf=i;
    }
    }
```

```
✓int n; 15
✓printf("Enter the number : ");
✓scanf("%d",&n);
✓int hf = 1;
for(int i=n/2;i>=1;i--){
    if(n%i==0){
        hf = i;
        ✓break;
    }
}
printf("Highest factor is %d : ",hf);
```



$i = 7, 6, 5$

1, 3, 5, 15

COLLEGE
WALLAH

Ques : WAP to check if a number is prime or not.

↓
write a
program

Any number 'n' is prime if
it has no factors except 1 & n

i = 2 to n-1

if (n % i == 0)

Q₁, WAP to check if no. is composite or not.

n = 24 , i = 2 , n/i

Continue;



'skip a round in the loop'

COLLEGE
WALLAH

Ques : WAP to print ^{even} ~~odd~~ numbers from 1 to 100.
Method-3 : Using continue

COLLEGE
WALLAH

While Loop

```
int i = 0;
while(i < 10) {
    // code
    i++;
}
```

→ condition ←

→ increment

```
for(int i = 0; i < 10; i++) {
    // code
}
```

COLLEGE
WALLAH

Q, Print no-s from 1 to 10 using while loop.

COLLEGE
WALLAH

Do- While Loop

→ runs atleast once

```
do {
```

```
    //code
```

```
} while ( another == 'y' );
```

Predict the output

```
main() {
```

```
    ✓int j;
```



```
    ✓while ( j <= 10 ) {
```

```
        printf ( "\n%d", j );
```

```
        j = j + 1; ⇒ j++;
```

```
    }
```

```
}
```

~~Print~~ → Garbage Values

Output

Predict the output

```
main() {
    ✓ int i = 1;
    ✓ while ( i <= 10 );
    {
        printf ( "\n%d", i );
        i++;
    }
}
```

1
i

Output

• 1
• 2
• 3
• 4
• 5
• 6
• 7
• 8
• 9
• 10

wrong

Predict the output

```
main() {
```

```
    ✓ int x = 1;
```

```
    ✓ while ( x == 1 ) {
```

```
        x = x - 1;
```

```
        printf ( "\n%d", x );
```

```
    }
```

```
}
```

0
1
x

Output

•
+ 0

Predict the output

```
main() {
```

```
    ✓ int x = 1;
```

```
    ✓ while ( x == 1 )
```

```
        x = x - 1;
```

```
    printf ( "\n%d", x );
```

```
}
```



Output

```
0
```

COLLEGE
WALLAH

Predict the output

```
main() {
```

```
    ✓ int x = 4, y = 3, z;
```

```
    ✓ z = 4x3-- - y;
```

```
        printf ( "\n%d %d %d", x, y, z );
```

```
}
```

$x--$ $\Rightarrow$ post decrement

↓

$x = x - 1$

$\begin{matrix} 3 \\ 4 \\ x \end{matrix}$

$\begin{matrix} 3 \\ y \end{matrix}$

$\begin{matrix} 1 \\ z \end{matrix}$

Output

.
3 3 1

Predict the output

```
main() {
    while ( 'a' < 'b' )
        printf ( "\nmalyalam is a palindrome" );
}
```

↓
m a l a y a l a m

'Infinite Loop'

COLLEGE
WALLAH

Predict the output

```
main() {
    ✓int i;
    ✓while ( i = 10 ) {
        printf ( "\n%d", i );
        i = i + 1;
    }
}
```



-
- 10
- 10
- 10
-

HW : Predict the output

```
main() {  
    int i = 10 ;  
    while ( i = 20 )  
        printf ( "\nA computer buff!" ) ;  
}
```

COLLEGE
WALLAH

Predict the output

```
main() {
```

```
    ✓int x = 4, y = 0, z;
```

```
    ✓while ( x >= 0 ) {
```

```
        x--;
```

```
        y++;
```

```
        if ( x == y )
```

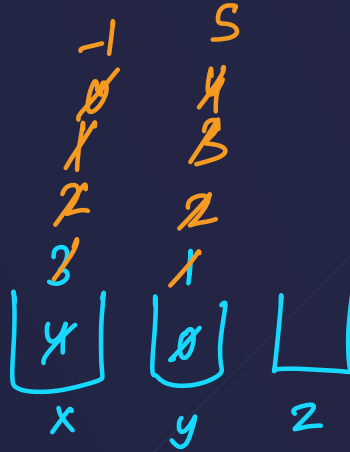
```
            continue;
```

```
        else
```

```
            printf ( "\n%d %d", x, y );
```

```
    }
```

```
}
```



Output

-
- 3 1
- 1 3
- 0 4
- -1 5

HW : Predict the output

```
main() {
    int x = 4, y = 0, z ;
    while ( x >= 0 ) {
        if ( x == y )
            break ;
        else
            printf ( "\n%d %d", x, y ) ;
            x-- ;
            y++ ;
    }
}
```

COLLEGE
WALLAH

Questions using Operators

$+$ $-$ $*$ $/$ $\%$

COLLEGE
WALLAH

Ques : WAP to count digits of a given number.

$n = 12458$

↓

5 digit no.

Algorithm :

n	count
12458	0
1245	1
124	2
12	3
1	4
0	5

Hint : continuously divide n by 10 untill it reaches 0.

Condition $\rightarrow (n > 0) / (n \neq 0)$

work $\rightarrow$ $n = n/10;$ $\rightarrow$ increment/
count++; decrement

Ques : WAP to print sum of digits of a given number.

$n = 12345$

#Hint 1: $sum = 1+2+3+4+5 = 5+4+3+2+1$

#Hint 2: If we want the last digit of a number 'n' $\rightarrow n\%10$

Algo

n	ld	sum
1234	4	0
123	3	4
12	2	7
1	1	9
0		10

khatam

HW 1: WAP to print sum of all the even digits of a given number.

$$n = 241312536$$

↓

$$\text{Sum} = 2 + 4 + 2 + 6$$

HW.2 WAP to print the product of all digits of a given number

COLLEGE
WALLAH

* **Ques** : WAP to print reverse of a given number.

$n = 1234$

$rn = 4321$

Hint - 1 : last digit

: $n = n/10$

Algorithm :

n	r	ld
1234	0	4
123	4	3
12	43	2
1	432	1
0	4321	

$r = r * 10;$

$r = r + ld;$

✓

HW : WAP to print the sum of given number and its reverse.

COLLEGE
WALLAH

Ques : Print the factorial of a given number 'n'.

$$\rightarrow 5! \text{ or } 5 = 5 \times 4 \times 3 \times 2 \times 1$$

$$\rightarrow 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$\Rightarrow 2! = 2 \times 1$$

$$\Rightarrow n! = n \times n-1 \times n-2 \times n-3 \dots 3 \times 2 \times 1$$

$$n! = \text{product of } 1, 2, 3, \dots n$$

$$\text{product} = 1;$$

HW : Print the factorials of first 'n' numbers

$n = 5$

↓

1

2

6

24

120

COLLEGE
WALLAH

HW : Write a program to print all the ASCII values and their equivalent characters of 26 alphabets using a while loop. (Capital)

Output : (Typecasting)

65 A

66 B

67 C

68 D

⋮

90 Z

COLLEGE
WALLAH

Auth to access the resources

Dev Tools

token

locker storage

break the authentication

can open
anywhere

2 mins to upload

list

3.11.3

data frame → to read excel file

offset self baari

COLLEGE
WALLAH

Chapter - 4

Loops-2

Pattern Printing Questions

* * * *

* * * *

* * * *

COLLEGE
WALLAH

Ques : Print the given pattern

```
*****
*****
*****
```

$m = \text{no. of rows / lines}$

$n = \text{no. of columns / no. of elements in each line}$

$m=2, n=3$

```
for(int i=1; i<=2m; i++){ // rows
    for(int j=1; j<=3n; j++){ // cols
        printf("*");
    }
    printf("\n");
}
```

Output

```
* * *
* * *
.
```

$i = 1, 2, 3$

$j = 1, 2, 3, 4, 1, 2, 3, 4$

Solid Rectangle

Ques : Print the given pattern

↓

```
****  
****  
****  
****
```

Input → 'n' as the side of square

Solid Square

COLLEGE
WALLAH

Ques : Print the given pattern

$n = 4$

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

$n = 3$

```
1 2 3
1 2 3
1 2 3
```

```
for(int i=1; i<=n; i++) {
    printf("%d", i);
}
```

```
1 2 3 4
```

$n = 4$

```
for(int i=1; i<=n; i++) {
    printf("*");
}
```

* * * *

Number Square

Homework :

Find out why

$n=4$

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4

```
for(int i=1;i<=n;i++){ // rows
    for(int j=1;j<=n;j++){ // cols
        printf("%d ",i);
    }
    printf("\n");
}
```

COLLEGE
WALLAH

Ques : Print the given pattern

$n \leq 26$

$n=4$

A B C D
A B C D
A B C D
A B C D

$n=2$

A B
A B

Hint : ASCII values

1 2 3

65 66 67

A B C

1 2 3

→

65 66 67 →

A B C

1 2 3

65 66 67

A B C

26 → Z

1 → A

2 → B

3 → C

& So on

Alphabet Square

Homework

$$n = 3$$

A A A

B B B

C C C

COLLEGE
WALLAH

Ques : Print the given pattern

$n = 4$

```
*
**
***
****
```

$n = 3$

```
x
x x
x x x
```

$n = 5$

	1	2	3	4	5	j
1	x					
2	x	x				
3	x	x	x			
4	x	x	x	x		
5	x	x	x	x	x	
i						

```
for(int i=1;i<=n;i++){ // rows
    for(int j=1;j<=i;j++){ // cols
        printf("* ");
    }
    printf("\n");
}
```

difference b/w
Square & triangle

Star Triangle

Ques : Print the given pattern

```
1
1 2
1 2 3
1 2 3 4
```

Number Triangle

COLLEGE
WALLAH

HW : Print the given pattern $n \leq 26$

```
A
A B
A B C
A B C D
```

Alphabet Triangle

COLLEGE
WALLAH

Homework : (Classwork)

1) For Ex = $n = 4$

1
2 2
3 3 3
4 4 4 4

2) $n = 4$

A
B B
C C C
D D D D

COLLEGE
WALLAH

*HW : Print the given pattern

$n = 5$

Hint : Use (if-else)

```
1
A B
1 2 3
A B C D
1 2 3 4 5
```

↳ Number Mix
Alphabet Triangle

COLLEGE
WALLAH

Ques : Print the given pattern

$n=4$

```

****
***
**
*
    
```

	1	2	3	4	→ j
1	*	*	*	*	
2	*	*	*		
3	*	*			
4	*				
i					

$(j=1; j \leq n+1-i; j++)$

$i=1, j=1 \text{ to } 4$

$i=2, j=1 \text{ to } 3$

$i=3, j=1 \text{ to } 2$

$i=4; i=1 \text{ to } 1$

$i+j = n+1$

↪ $j = n+1-i$

Star Triangle Ula

HW : Print the given pattern

```
1 2 3 4
1 2 3
1 2
1
```



```
1 1 1 1
2 2 2
3 3
4
```

Number Triangle Upta

COLLEGE
WALLAH

HW : Print the given pattern

```

A B C D
A B C
A B
A
  
```

&

```

A A A A
B B B
C C
D
  
```

Alphabet Triangle Ulta

COLLEGE
WALLAH

Ques: Odd Number Square

$n=4$

1 3 5 7

1 3 5 7

1 3 5 7

1 3 5 7

$n=3$

1 3 5

1 3 5

1 3 5

COLLEGE
WALLAH

Ques : Print the given pattern²

H.W.

```
1
1 3
1 3 5
1 3 5 7
```

△

```
1 3 5 7
1 3 5
1 3
1
```

Odd Number Triangle

COLLEGE
WALLAH

Homework : $n=3$ Ex

2 4 6

2 4 6

2 4 6

Even Number
Square

Δ

2

2 4

2 4 6

Even Number
triangle

Δ

2 4 6

2 4

2

Even Number
triangle Ueta

COLLEGE
WALLAH

***Ques** : Print the given pattern

$n = 4$

```

1
2 3
4 5 6
7 8 9 10
    
```

$n = 2$

```

1
2 3
    
```

$n = 5$

```

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
    
```

Homework :

```

1 2 3
4 5 6
7 8 9
    
```

```

1 2 3
4 5
6
    
```

Floyd's Triangle

***Ques** : Print the given pattern

$n=4$

```

1
01
101
0101
    
```

	1	2	3	4	5	→ j
1	1					
2	0	1				
3	1	0	1			
4	0	1	0	1		
5	1	0	1	0	1	
i						

if $(i+j \% 2 == 0)$ print (1)
else 0

0 & 1 Triangle

Ques : Print the given pattern

n is odd

n=5

```

--*--
--*--
*****
--*--
--*--
    
```

→

	1	2	3	4	5	j
1	#	#	*	#	#	
2	#	#	*	#	#	
3	*	*	*	*	*	
4	#	#	*	#	#	
5	#	#	*	#	#	
i						

*if (i == mid || j == mid) → **
else → #

n=5

int mid = $\frac{n}{2} + 1;$

HW : Print the given pattern

	1	2	3	4	5	6
1	*	*	*	*	*	*
2	*	_	_	_	_	*
3	*	_	_	_	_	*
4	*	*	*	*	*	*

Conditions for '*'

1) First row

2) last Row

3) First Col

4) Last Col

Hollow Rectangle

COLLEGE
WALLAH

HW : Print the given pattern

	1	2	3	4	5	j
1	*	-	-	-	*	
2	-	*	-	*	-	
3	-	-	*	-	-	
4	-	*	-	*	-	
5	*	-	-	-	*	
i						

when $i = j$ *
 $i + j = n + 1$

Star Cross

COLLEGE
WALLAH

***Ques** : Print the given pattern

	1	2	3	4	j
1	—	—	—	*	
2	—	—	*	*	
3	—	*	*	*	
4	*	*	*	*	
i					

$n=4$

→ if ($i+j \geq n+1$) *
 else —

Method-1

Star Triangle Reverse

COLLEGE
WALLAH

*Ques : Print the given pattern

Method-2

→ Using 2 loops inside loop

```

1 2 3 4
1  _ _ _ *
2  _ _ **
3  _ ***
4  ****
    
```

```

      1 2 3 4
      1  _ _ _ 3
      2  _ _ 2
      3  _ 1
      4  . 0
    
```

```

      1 2 3 4
      1 *
      2 * *
      3 * * *
      4 * * * *
    
```

→ $j \leq n-i$

$i+j=n$

Star Triangle Reverse

~~Ques~~ : Print the given pattern

H.W.

```

  _ _ _ 1
  _ _ 1 2
  _ 1 2 3
  1 2 3 4
  
```

Number Triangle Reverse

COLLEGE
WALLAH

HW : Print the given pattern

```
  A
 AB
ABC
ABCD
```

Alphabet Triangle Reverse

COLLEGE
WALLAH

HW : Print the given pattern



```
-----*****  
-----****  
-----***  
-----**
```

Rhombus

COLLEGE
WALLAH

*Ques : Print the given pattern

```

1 2 3 4 5 6 7
1  _ _ _ *
2  _ _ * * *
3  _ * * * * *
4 * * * * * *

```

→

```

1  _ _ _
2  _ _
3  _
4  .

```

+

```

1  *
2  * * *
3  * * * * *
4  * * * * * *

```

↓

$j \leq 2i^2 - 1$

Star Pyramid

COLLEGE
WALLAH

***Ques** : Print the given pattern

```

1  _ _ _ *
2  _ _ ***
3  _ ****
4  *****
  
```

$n=4$

$n=3$

```

1  _ _ x
2  _ x x x
3  x x x x x
  
```

$nsb \rightarrow n-1$ and $\boxed{-1}$

$nst \rightarrow 1$ and $\boxed{+2}$

Star Pyramid

COLLEGE
WALLAH

HW : Print the given pattern

```

      1
     1 2 3
    1 2 3 4 5
   1 2 3 4 5 6 7
  
```

Solve by

M1 & M2

Number Pyramid

COLLEGE
WALLAH

HW : Print the given pattern

```
  A
 A B C
A B C D E
A B C D E F G
```

Alphabet Pyramid

COLLEGE
WALLAH

***HW** : Print the given pattern

```
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

Number Pyramid Palindrome

COLLEGE
WALLAH

****Ques** : Print the given pattern

```

1  _ _ _ *
2  _ _ * * *
3  _ * * * * *
4  * * * * * *

```

```

1  _ * * * * *
2  _ _ * * *
3  _ _ _ *

```

$n=4$

$\rightarrow nsp = 1, +1$
 $nst = 2n - 3, -2$

Star Diamond

COLLEGE
WALLAH

**Ques : Print the given pattern

```

1      *
  _ _ _
2    _ _ * * *
3  _ _ * * * * *
4 * * * * * * *
5 _ _ * * * * *
6 _ _ * * *
7 _ _ _ *
i
n = 4
  
```

$nsp = n - 1$

$nst = 1$

```

if(i < n){
    |   nsp--;
      |   nst += 2;
3
do { // i >= n
    |   nsp++;
      |   nst -= 2;
3
  
```

Star Diamond

****Ques** : Print the given pattern

```

      *
     ***
    *****
   *****
  *****
 *****
  *****
   *****
    *****
     *****
      *
  
```

$n=1$

```

*
  
```

$n=2$

```

  *
 * *
* * *
  *
  
```

$n=3$

```

    *
   * *
  * * *
 * * * *
* * * * *
 * * * *
  * * *
   * *
    *
  
```

Star Diamond

COLLEGE
WALLAH

HW : Print the given pattern

```

      * * * * *
    1 * * * _ * * *
    2 * * _ _ _ * *
    3 * _ _ _ _ *
  
```

$n=4$

=

```

      * * *      _      * * *
      * *      +   _ _ _      *
      *              _ _ _ _ _      *
  
```

***HW : Print the given pattern**

```

1 2 3 4 5 6 7
1 2 3   5 6 7
1 2     6 7
1       7
    
```

Number Bridge

COLLEGE
WALLAH

***HW** : Print the given pattern

```

A B C D E F G
A B C   E F G
A B       F G
A           G
    
```

Alphabet Bridge

COLLEGE
WALLAH

****HW : Print the given pattern**

```

4 4 4 4 4 4 4
4 3 3 3 3 3 4
4 3 2 2 2 3 4
4 3 2 1 2 3 4
4 3 2 2 2 3 4
4 3 3 3 3 3 4
4 4 4 4 4 4 4
    
```

Number Pyramid Mast

COLLEGE
WALLAH

Ch:5

Functions

& Pointers

COLLEGE
WALLAH

What and Why?

input $\rightarrow$ f $\rightarrow$ Output

$$y = f(x) = x^2 + 5 \Rightarrow f(8) = 8^2 + 5 = 69$$

Repetitions $\rightarrow$ Loops $\rightarrow$ for things that are together

Readable $\rightarrow$ After some time

Basic syntax

```
fun(){
```

```
// code
```

```
}
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
int main(){
```

```
}
```

→ main
function

COLLEGE
WALLAH


```

1  #include<stdio.h>
2  void greet(){
3      printf("Hey, Good Morning\n");
4      printf("How are you?\n");
5  }
6  int main(){
7      printf("Hi, I am Raghav Garg\n");
8      greet(); // calling the function
9      printf("Hi, I work at PW\n");
10     return 0;

```

return type

1, 6, 7, 8, 2, 3, 4, 5, 9, 10

Output

- Hi, I am Raghav Garg
- Hey, Good Morning
- How are you ?
- Hi, I work at PW
-

```
main()  
return ;
```

↓

- 1) Khatam
- 2) Give the value (if any) back to calling function

Parameters

arguments, parameters

```

✓ #include <stdio.h>
void oddEven(int x){
    if(x%2==0) printf("%d is even",x);
    else printf("%d is odd",x);
}
✓ int main(){
    ✓ int a;
    ✓ printf("Enter a number : ");
    ✓ scanf("%d",&a);
    ✓ oddEven(a); // Pass by Value
    return 0;
}
    
```

6
x

• Enter a number : 6

• 6 is Even

6
a

Return type :

return means finish &
give back the value

```

✓ #include <stdio.h>
✓ int min(int x, int y){
  ✓ if(x < y) return x;
  ✓ else return y;
}

✓ int main(){
  ✓ int a;
  ✓ printf("Enter 1st number : ");
  ✓ scanf("%d", &a);
  ✓ int b;
  ✓ printf("Enter 2nd number : ");
  ✓ scanf("%d", &b);
  ✓ int m = min(6, 2);
  ✓ printf("Minimum of %d and %d is %d", a, b, m);
  return 0;
}
  
```

$\boxed{6}$ $\boxed{2}$
 x y

$\boxed{6}$ $\boxed{2}$
 a b

$\boxed{2}$
 m

Enter 1st no: 6
Enter 2nd no: 2

How functions work : ek ke andar doosra, doosre ke andar teesra

```

✓ #include <stdio.h>
✓ void japan(){
  ✓ printf("You are in Japan\n");
}
✓ void england(){
  ✓ printf("You are in England\n");
  ✓ japan(); ●
}
✓ void india(){
  ✓ printf("You are in India\n");
  ✓ england(); ●
}
✓ int main(){
  ✓ printf("You are in main\n");
  ✓ india(); ●
}
  
```

- main
- India
- England
- Japan

Function Prototype – useless



To counter the problem of calling

```
int fun(int a int b);
```

Kaam ki baate :

- 1) `main()` ek hi baar aata hai.
- 2) Starts with `main`
- 3) unlimited functions

COLLEGE
WALLAH

Return type : Sum function se samajhte hai

```
int sum(int a, int b){  
    | return a+b;  
}
```

```
int main(){  
    | printf("%d", sum(8,7));  
}
```


Library functions

↪ $\text{sqrt}(16) \rightarrow 4$

↪ $\text{pow}(2, 5) \rightarrow 32$

COLLEGE
WALLAH

Library functions

COLLEGE
WALLAH

```
✓ #include<stdio.h>
  int min(int a, int b){
    if(a<b) return a; ✓
    else return b;
  }
```

```
✓ int main(){
  ✓ int a = 3;
  ✓ int b = 9;
  ✓ int m = min(a,b);
  ✓ printf("Minimum of %d and %d is %d",a,b,m);
  return 0;
}
```



Ques : Combination and Permutation

→ Factorial : $n! = n \times n-1 \times n-2 \times \dots \times 3 \times 2 \times 1$

Combinations : ${}^nC_r = \frac{n!}{r! \times (n-r)!}$ | ${}^5C_2 = \frac{5!}{2! \times 3!} = \frac{5 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times 1}{2 \times 1 \times \cancel{3} \times \cancel{2} \times 1} = 10$

Permutation : ${}^nP_r = \frac{n!}{(n-r)!}$ → Homework

fun → combi(n, r) → nC_r

Ques : Combination and Permutation

$${}^nC_0 = \frac{n!}{0!(n-0)!} = \frac{\cancel{n!}}{1 \cdot \cancel{n!}} = 1$$

$${}^nC_1 = \frac{n!}{1!(n-1)!} = \frac{n \times \cancel{(n-1)!}}{\cancel{(n-1)!}} = n$$

Ques : Pascal triangle $\rightarrow n=5$

$$\begin{array}{cccccc}
 & & & & 1 & \\
 & & & 1 & & 1 \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 4 & & 6 & & 4 & & 1 \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1
 \end{array}$$

$$\begin{array}{ccccccccc}
 & & & & & & 1 & & \\
 & & & & 1 & & 1 & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
 1 & & 4 & & 6 & & 4 & & 1 \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1
 \end{array}$$

$0! = 1, 1! = 1$

Ques : Pascal triangle

	0	1	2	3	4	5
0	1					
1	1	1				
2	1	2	1			
3	1	3	3	1		
4	1	4	6	4	1	
5	1	5	10	10	5	1

	0	1	2	3	4	5
0	0C_0					
1	1C_0	1C_1				
2	2C_0	2C_1	2C_2			
3	3C_0	3C_1	3C_2	3C_3		
4	4C_0	4C_1	4C_2	4C_3	4C_4	
5	5C_0	5C_1	5C_2	5C_3	5C_4	5C_5

row no. i C_j $\rightarrow$ col no.

Ques : Pascal triangle

Homework

```

-- -- 1
-- -- 1 -- 1
-- 1 -- 2 -- 1
-- 1 -- 3 -- 3 -- 1
1 -- 4 -- 6 -- 4 -- 1
    
```

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Scope of variable

Local & Global

AuKat

```
✓int a = 7;
✓void change(){
    a = 5;
}
✓int main(){
    ✓int a = 70;
    ✓printf("%d ",a);
    ✓change();
    ✓printf("%d ",a);
```

5
7
a

70 70

70
a

Formal parameters and Actual Parameters

```
#include<stdio.h>
int min(int x, int y){ → formal parameters
    if(x<y) return x;
    else return y;
}
int main(){
    int a = 3;
    int b = 9;
    int m = min(a,b); → Actual Parameters
    printf("Minimum of %d and %d is %d",a,b,m);
    return 0;
}
```

FP
↓
variables/
containers

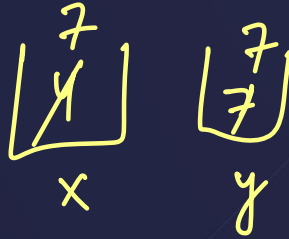
AP
↓
the actual
values

Ques : Swap 2 numbers

```

✓ int x;
✓ scanf("%d",&x);
✓ int y;
✓ scanf("%d",&y);
✓ printf("%d %d\n",x,y);
// swap
✓ x = y;
✓ y = x;
✓ printf("%d %d\n",x,y);
return 0;

```



x = y; x is changed now permanently changed

Output

4
7

4 7

7 7

Ques : Swap 2 numbers

M-I : Using extra dabba / variable →

M-II : Without using 3rd variable

```
int temp = x;
x = y;
y = temp;
```

Swap (x, y)

↓

Algorithm :

$$x = x + y$$

$$y = x - y$$

$$x = x - y$$

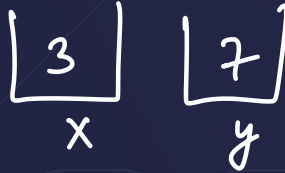
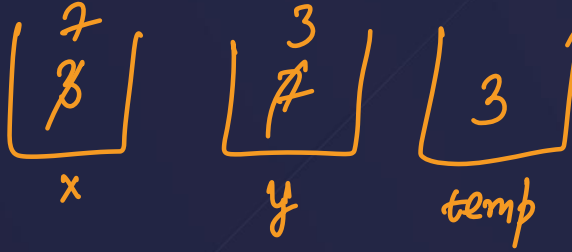
$\begin{array}{r} 6+2-(6+2-2) \\ \underline{6+2} \\ 6 \end{array}$	$\begin{array}{r} 6+2-2 \\ \underline{7} \end{array}$
x	y

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Pass by value & Pass by reference

```
#include<stdio.h>
void swap(int x, int y){
    int temp = x;
    x = y;
    y = temp;
}

int main(){
    int x;
    scanf("%d",&x);
    int y;
    scanf("%d",&y);
    printf("%d %d\n",x,y);
    //swap
    swap(x,y);
    printf("%d %d\n",x,y);
    return 0;
}
```



Output/ Input

3

7

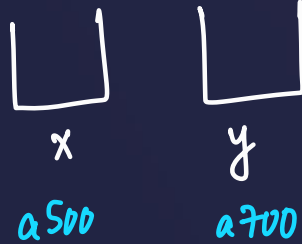
3 7

3 7

*Pointers

int x;

int y;

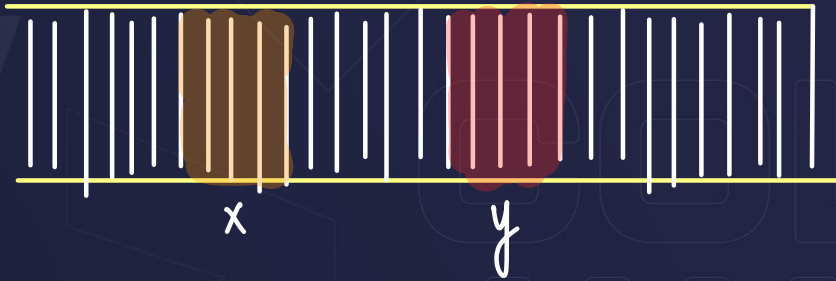


int → 4 bytes

float → 4 bytes

char → 2 bytes

bool → 1 byte



*Pointers

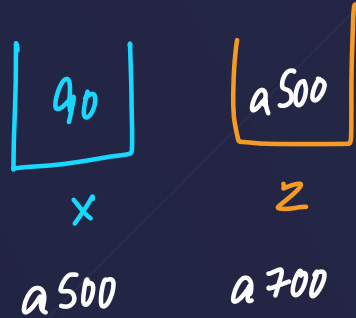
```

✓ int x = 90;
✓ int* z = &x;
✓ printf("%d\n", x);
✓ printf("%p\n", z);
✓ printf("%d\n", *z);

```

* → dereference
operator

Star operator



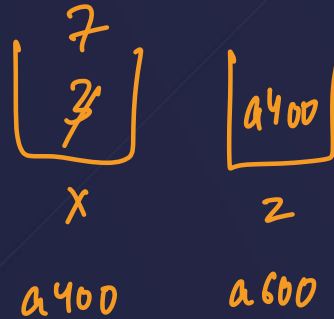
Output

- 90
- a500
- 90

*Pointers

```

✓ int x = 3;
✓ int* z = &x;
✓ printf("%d\n", x);
✓ *z = 7;  Update/change
✓ printf("%d\n", x);
    
```



• 3
• 7

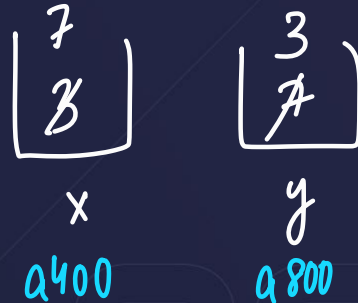
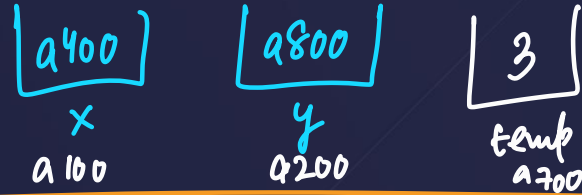
*Pointers

→ Swap 2 numbers

swap1 ;

Output

```
void swap(int* x, int* y){
    int temp = *x;
    *x = *y;
    *y = temp;
}
```



Pass by Reference

3
7
3 7

```
int main(){
    printf("\n");

    int x;
    scanf("%d",&x);
    int y;
    scanf("%d",&y);
    printf("%d %d\n",x,y);
    swap(&x,&y);
    printf("%d %d\n",x,y);

    printf("\n");
    return 0;
}
```

State TRUE or FALSE :

- 1) The same variable names can be used in different functions without any conflict. *TRUE*

State TRUE or FALSE :

- 2) Every called function must contain a return statement. *FALSE* (exception is void)
- 3) A function may contain more than one return statements. *TRUE*



but only one return statement is actually executed

State TRUE or FALSE :

4) Same names can be used for different functions without any conflict. FALSE

↓
C++ & Java are OOP Languages
but C is not

State TRUE or FALSE :

5) A function may be called more than once from any other function **TRUE**

6) A function can call itself (except main) **TRUE**

ek badi problem →

using a chhoti problem
↓
combinations

Easy

Recursion



a function calling itself

Recursive Relations

Function calling itself

```
✓ #include<stdio.h>
✓ void greet(){
    ✓ printf("Good Morning\n");
    ✓ greet();
}
✓ int main(){
    ✓ greet();
    return 0;
}
```

- Good Morning
- Good Morning

How to stop it?

```

void greet(int2 n){
    ✓ if(n=0) return;
    ✓ printf("Good Morning\n");
    ✓ greet(n1-1);
}
int main(){
    ✓ greet(2);
    ✓ return 0;
}
    
```

```

void greet(int1 n){
    ✓ if(n=0) return;
    ✓ printf("Good Morning\n");
    ✓ greet(n0-1);
}
    
```

```

void greet(int0 n){
    ✓ if(n==0) return;
    ✗ printf("Good Morning\n");
    ✗ greet(n-1);
}
    
```

Comprehensively
DRY RUN

Output

- Good Morning
- Good Morning

Ques : Print n to 1 (using recursion)

For Ex : $n = 5 \rightarrow$ Output

↓

5

4

3

2

1

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Ques : Print 1 to n → Kind of harder than n to 1

M-I: Using extra parameter

Output

- . 1
- . 2
- . 3
- .

```
void print(int1 x, int3 n){
    ✓ if(x>n) return; // base case
    ✓ printf("%d\n",x); // kaam
    ✓ print(x+1,n); // call
}
```

```
void print(int4 x, int3 n){
    ✓ if(x>n) return; // base case
    printf("%d\n",x); // kaam
    print(x+1,n); // call
}
```

```
void print(int2 x, int3 n){
    ✓ if(x>n) return; // base case
    ✓ printf("%d\n",x); // kaam
    ✓ print(x+1,n); // call
}
```

```
void print(int3 x, int3 n){
    ✓ if(x>n) return; // base case
    ✓ printf("%d\n",x); // kaam
    ✓ print(x+1,n); // call
}
```

Ques : Print 1 to n (after recursive call)

→ Without using the extra parameter

```
void print(int2 n){
    ✓ if(n==0) return; // base case
    ✓ print(n1); // call
    ✓ printf("%d\n", n); // kaam
}
```

```
void print(int1 n){
    ✓ if(n==0) return; // base case
    ✓ print(n0); // call
    ✓ printf("%d\n", n); // kaam
}
```

```
void print(int3 n){
    ✓ if(n==0) return; // base case
    ✓ print(n2); // call
    ✓ printf("%d\n", n); // kaam
}

int main(){
    ✓ int n;
    ✓ printf("Enter n : ");
    ✓ scanf("%d", &n);
    ✓ print(3);
    ✓ return 0;
}
```

```
void print(int0 n){
    ✓ if(n==0) return; // base case
    print(n-1); // call
    ✓ printf("%d\n", n); // kaam
}
```

Output

Enter n : 3

- 1
- 2
- 3

Concept in Recursion which is VVVIMP:

```
fun() {  
    | base case  
    | work  
    | fun();  
    | work  
    |  
    3  
}
```

Ques : Print sum from 1 to n (Parameterised)

```
void sum1toN(int3 n, int4 sum){
    if(n==0)α{
        printf("%d",sum);
        return;
    }
    sum1toN(n-1,sum+n); // call
}
```

```
void sum1toN(int2 n, int7 sum){
    if(n==0)α{
        printf("%d",sum);
        return;
    }
    sum1toN(n-1,sum+n); // call
}
```

```
void sum1toN(int4 n, int0 sum){
    if(n==0){
        printf("%d",sum);
        return;
    }
    sum1toN(n-1,sum+n); // call
}
```

```
void sum1toN(int1 n, int9 sum){
    if(n==0)α{
        printf("%d",sum);
        return;
    }
    sum1toN(n-1,sum+n); // call
}
```

```
int main(){
    int n;
    printf("Enter n : ");
    scanf("%d",&n);
    sum1toN(4n,0);
    return 0;
}
```

```
void sum1toN(int0 n, int10 sum){
    if(0==0)α{
        printf("%d",sum);
        return;
    }
    sum1toN(n-1,sum+n); // call
}
```

Output

Enter n: 4.

10

Ques : Print sum from 1 to n (Return type)

→ void, int ✓

↓
n to 1

$$\text{Sum}(5) = 5 + 4 + 3 + 2 + 1$$

$$\text{Sum}(4) = 4 + \boxed{3 + 2 + 1}$$

$$\text{Sum}(3) = 3 + 2 + 1$$

$$\text{Sum}(2) = 2 + 1$$

$$\text{Sum}(1) = 1$$

$$\text{Sum}(5) = 5 + \text{Sum}(4)$$

$$\text{Sum}(4) = 4 + \text{Sum}(3)$$

$$\text{Sum}(3) = 3 + \text{Sum}(2)$$

$$\text{Sum}(2) = 2 + \text{Sum}(1)$$

$$\text{Sum}(1) = 1$$

Ques : Print sum from 1 to n (Return type)

$$\overset{15}{\text{sum}(5)} = 5 + \overset{10}{\text{sum}(4)}$$

$$\downarrow \nearrow \overset{6}{4 + \text{sum}(3)}$$

$$\downarrow \nearrow \overset{3}{3 + \text{sum}(2)}$$

$$\downarrow \nearrow \overset{1}{2 + \text{sum}(1)}$$

$$\downarrow \nearrow 1$$

Recursive tree / Ladder

recursive
formula

$$\text{Sum}(n) = n + \text{sum}(n-1)$$

recurrence relation

$$\text{Sum}(1) = 1$$

Ques : Make a function which calculates the factorial of n using recursion.

$$n! = n \times \boxed{n-1 \times \dots \times 3 \times 2 \times 1}$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$\text{fact}(5) = 5 \times \boxed{4 \times 3 \times 2 \times 1}$$

$$\text{fact}(4) = 4 \times 3 \times 2 \times 1$$

$$\text{fact}(5) = 5 \times \text{fact}(4)$$

$$\boxed{\text{fact}(n) = n \times \text{fact}(n-1)}$$

$$\text{fact}(1) = 1$$

$$\text{fact}(0) = 1$$

Ques : Make a function which calculates 'a' raised to the power 'b' using recursion.

$$a^b$$

↓

$$2^5 = 32$$

$$a^{(4)} = a \times a \times a \times a$$

$$a^2 = a \times a$$

$$2^5 = 2 \times 2 \times 2 \times 2 \times 2$$

$$a^b = \underbrace{a \times a \times a \times a \dots a}_{b \text{ terms}}$$

$$a^b = a \times (a \times a \times a \dots)$$

b-1 times

$$a^b = a \times a^{b-1}$$

$$2^4 = 2 \times (2 \times 2 \times 2)$$

$$2^4 = 2 \times 2^3$$

$$a^0 = 1$$

$$0^0 \rightarrow x$$

if (b==0) return 1

↙

$$** \text{power}(a, b) = a * \text{power}(a, b-1)$$

*Multiple Calls

Ques : Write a function to calculate the nth fibonacci number using recursion.

	1	1	2	3	5	8	13	21	34	55	89	144	2 co on
n =	1	2	3	4	5	6	7	8	9	10	11	12	

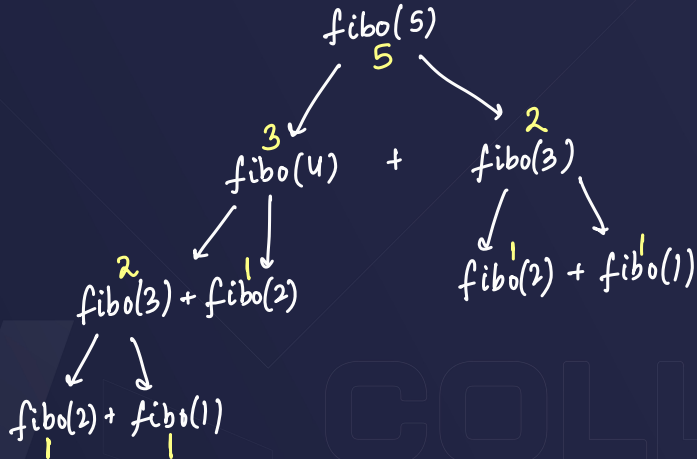
$$\text{fibonacci}(7) = \text{fibonacci}(6) + \text{fibonacci}(5)$$

$$\text{fibonacci}(n) = \text{fibonacci}(n-1) + \text{fibonacci}(n-2)$$

recurrence relation

base : if $n == 1$ or $n == 2$ return 1

Recursion Tree



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Playlists



C Programming



Recursion

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HW : Power function (logarithmic)

↓
 $a^b = ?$

$$\begin{aligned} 3^8 &= 3 \times 3^7 \\ 3^7 &= 3 \times 3^6 \\ 3^6 &= 3 \times 3^5 \\ 3^5 &= 3 \times 3^4 \\ 3^4 &= 3 \times 3^3 \\ 3^3 &= 3 \times 3^2 \\ 3^2 &= 3 \times 3^1 \\ 3^1 &= 3 \times 3^0 \end{aligned}$$

$$\begin{aligned} 3^{32} &= 3 \times 3^{31} \\ 3^{31} &= 3 \times 3^{30} \\ &\vdots \\ 3^1 &= 3 \times 3^0 \end{aligned}$$

1

32
calls

$$\begin{aligned} 3^{32} &= 3^{16} \times 3^{16} \\ 3^{16} &= 3^8 \times 3^8 \\ 3^8 &= 3^4 \times 3^4 \\ 3^4 &= 3^2 \times 3^2 \\ 3^2 &= 3^1 \times 3^1 \\ 3^1 &= 3 \end{aligned}$$

6 steps
calls

int x; 

Arrays

```
int m1 = 80;
```

```
int m2 = 45;
```

```
⋮
```

```
int m10 = 77;
```

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What is an array?



It is can be said as a list of items

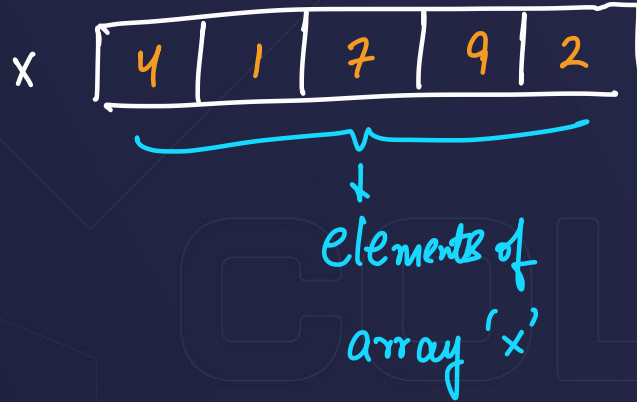


int / float / char / bool

Syntax and Declaration

arr

`int x[5] = {4, 1, 7, 9, 2};`
 [↓
 data size
 type



Accessing Elements of Array

→ using square brackets & indexing

Updation

```
int arr[4] = { 10, 20, 30, 40 };
```

arr

0	1	2	3
10	20	30	40

→ `printf("%d", arr[2]);`

Printing Output and Taking Input

↓
If I have to print all the elements
of an array
↓
Loops

```
int x;  
scanf("%d", &x);
```

→ If array is of size 'n' then the indexing
will be from 0 to n-1

```
int arr[] = {10, 20, 30, 40};
```



↓
16 bytes

```
arr[2] = 90
```

↓
`int x;` → 4 bytes

in

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Ques : Are the following array declarations correct?

`int a (25);` *Wrong* → `int a[25];`

`int size = 10, b[size];` *Correct*

`int c = {0,1,2};` *wrong*

→ `int size = 10;`
`int b[size];`

↓
`int c[] = {0,1,2};` ✓

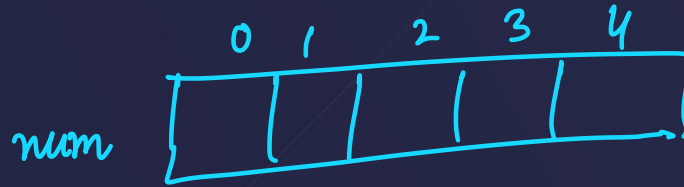
`int c[3] = {0,1,2};` ✓

Ques : Which element of the array does this expression reference?

num[4]



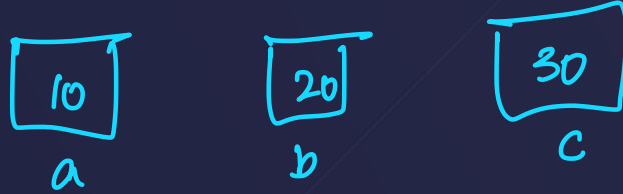
5th element from
start



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Memory Allocation in Arrays

```
int a = 10;
int b = 20;
int c = 30;
↓
```



```
int arr[3] = {10, 20, 30};
```

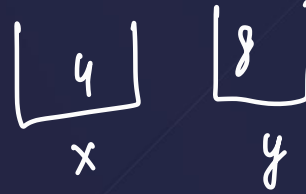
arr

0	1	2
10	20	30

↳ Contiguous memory allocation

Passing Array to Functions

```
void swap(int x, int y){
|
```



```
3
-----
int main(){
```

```
|   int x=4;
|   int y=8;
|   swap(x,y);
|
3
```



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Passing Array to Functions

↓
for int, float, char → pass by value → pointers → pass by reference
↘ different behaviour

Ques : Calculate the sum of all the elements in the given array.

```
int arr[] = { 5, 4, 1, 9, 6 };
```

Sum = 0; 5 9 10 19 25

Homework : Calculate the product of all the elements in the given array.

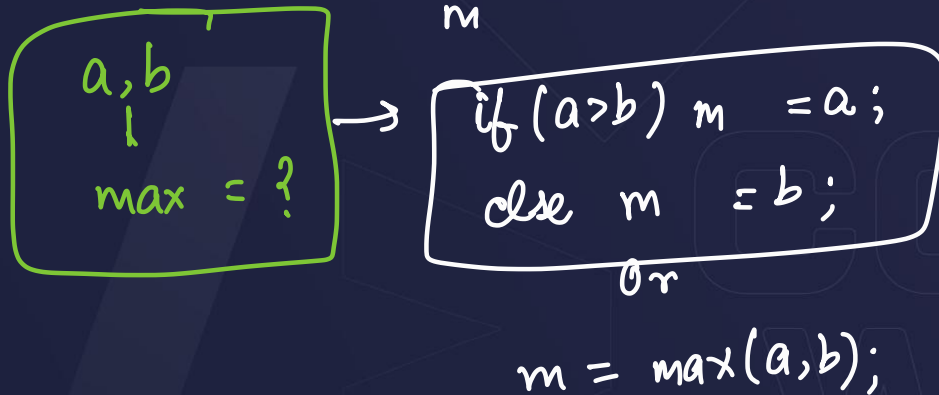
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Ques : Find the maximum value out of all the elements in the array.

0
1
2
3
4

`int arr[5] = { 4, 8, 1, 9, 3 } ;`
`mx = 4; 8 9`

`int mx = arr[0];`



Ques : Find the maximum value out of all the elements in the array.

Approach-2 : `int arr[4] = { 4, 8, 1, 6 } ;`

↓
`int mx = -1;` → problem

Tyada Use hote → `int mx = INT_MIN;`

Homework : Find the minimum value out of all the elements in the array.

↓

2 methods

↓

`int mn = arr[0];` & then compare with 1 to n-1

→ `int mn = INT_MAX;`

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MCQ : What is the difference between the 5's in these two expressions?

↗ size
int num[5]; → declaration

num[5] = 11;
index ↘ update, assign, initialise

1. first is particular element, second is type
- ✓ 2. first is array size, second is particular element/index
3. first is particular element, second is array size
4. both specify array size

State TRUE or FALSE :

1. The array `int num[26]` has twenty-six elements. *True*
2. The expression `num[1]` designates the first element in the array *FALSE*
3. It is necessary to initialize the array at the time of declaration. *FALSE* `int arr[5];`
4. The expression `num[27]` designates the twenty-eighth element in the array. *TRUE*

HW : Given an integer n . Create an array containing squares of all natural numbers till n and print the elements of the array.

$$n = 5$$

1	4	9	16	25
---	---	---	----	----

Ques : Given an array of integers, change the value of all odd indexed elements to its second multiple and increment all even indexed value by 10.



Ques : Count the number of elements in given array greater than a given number x.

$arr = \{ 1, 2, 3, 4, 5, 6, 7 \}; \quad target = 4$

HW : Find the difference between the sum of elements at even indices to the sum of elements at odd indices.

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Ques: Search for an element in an array

arr = { 1, 2, 4, 8, 16, 12, 9, 3 } x = 8

bool → true/false
↓ 1 / 0

bool flag = false;

Ques : Find the second largest element in the given Array. *→ distinct number*

arr = { 4, 1, 2, 2, 5, 6, 6, 8, 9, 9 }

Hint : 2 times for loop

→ int max, smax

Ques : Write a program to copy the contents of one array into another in the reverse order.

$$\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \text{arr} = & \{ 1, 2, 3, 4, 5, 6, 7 \} \end{array}$$

$n=7$

$\text{brr}[i] = \text{arr}[n-1-i]$

$$\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \text{brr} = & \{ 7, 6, 5, 4, 3, 2, 1 \} \end{array}$$

$\text{brr}[0] = 7 = \text{arr}[6]$

$\text{brr}[1] = \text{arr}[5]$

$\text{brr}[2] = \text{arr}[4]$

$\text{brr}[i] = \text{arr}[j]$

$\Rightarrow i+j = n-1$

$\Rightarrow j = n-1-i$

Ques : Write a program to reverse the array without using any extra array.

$arr = \{ 1, 2, 3, 4, 5, 6, 7 \}$


 $arr = \{ \begin{matrix} & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 7, & 6, & 5, & 4, & 3, & 2, & 1 \end{matrix} \}$

 i

 j

```
int i = 0;
int j = n - 1;
```

```
while (i < j) {
    swap(arr[i], arr[j])
    i++;
    j--;
}
```

Homework : If an array `arr` contains `n` elements, then check if the given array is a palindrome or not.

→ `racecar`

→ `malayalam`

→ `arr = { 1, 2, 3, 4, 3, 2, 1 }`

`arr = { 1, 2, 2, 1 }`

1D array / Array $\rightarrow$ list



5 students $\rightarrow$ P, C, M

2D Arrays



2 dimensional arrays

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What and Why?

```
int arr[n] ;  
int arr[r][c];
```

So far we have explored arrays with only one dimension. It is also possible for arrays to have two or more dimensions. The two dimensional array is also called a matrix.

```
int arr[r][c];
```

This is a 2D array where **r** depicts number of rows in matrix and **c** depicts number of columns in the matrix.

```
int arr[3][3]
```


```
int arr[2][2];
```

	0	1
0	1	2
1	3	4

```
int arr[2][4];
```


```
arr[0][0] = 1
```

```
arr[0][1] = 2
```

```
arr[1][0] = 3
```

```
arr[1][1] = 4
```

2D arrays can be called as array of 'arrays'

$$\left. \begin{array}{l} \rightarrow \{1, 2, 3\} \\ \rightarrow \{4, 5, 6\} \\ \rightarrow \{7, 8, 9\} \end{array} \right\} \rightarrow \{ \{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\} \}$$

Initialisation of a 2-Dimensional Array

4 rows 2 cols
 ↑ ↑

```
int arr[4][2] = { { 1234, 56 }, { 1256, 43 }, { 1434, 32 }, { 1312, 96 } } ;
```

```
int arr[4][2] = { 1234, 56 , 1256, 43 , 1434, 32 , 1312, 96 } ;
```

```
int arr[2][3] = { 12, 34, 56, 78, 91, 23 } ;
```

```
int arr[ ][3] = {12, 34, 56, 78, 91, 23 } ; → This works
```

```
int arr[2][ ] = { 12, 34, 56, 78, 91, 23 } → ✗
```

`int arr[4][2] = { {1, 2}, {3, 4}, {5, 6}, {7, 8} };`

↓

	0	1
0	1	2
1	3	4
2	5	6
3	7	8

or

`int arr[4][2] = { 1, 2, 3, 4, 5, 6, 7, 8 };`

Ques : Write a program to store roll number and marks obtained by 4 students side by side in a matrix.

Ragharv	Harsh	Urvi	Sanket
76	88	82	71
80	82	91	98

→

Ragharv	76	80
Harsh	88	82
Urvi	82	91
Sanket	71	98

`int arr[4][2]`

Output / Print : Nested loop

	0	1	2	→ j
0	1	2	3	
1	4	5	6	
2	7	8	9	

↓
i



Output

1 2 3
4 5 6
7 8 9

1, 2, 3 → arr[0][2]
↓
arr[0][0] arr[0][1]

Output → Column Wise

→ Transpose

1	2	3
4	5	6
7	8	9

Output

1 4 7
2 5 8
3 6 9

Ques : Write a program to store 10 at every index of a 2D matrix with 5 rows and 5 columns.

10	10	10	10	10
10	10	10	10	10
10	10	10	10	10
10	10	10	10	10
10	10	10	10	10

H.W.

~~Ques~~ : Write a program to add two matrices.

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline 4 & 3 & 1 \\ \hline 6 & 5 & 2 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline 5 & 5 & 4 \\ \hline 10 & 10 & 8 \\ \hline \end{array}$$

a
 b
 c

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Ques : Find the maximum element in a given matrix.

↓
Same as 1D array

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HW : Find the minimum element in a given matrix.

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HW : Find the sum of all elements in a given matrix.

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HW : Find the product of all elements in a given matrix.

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HW : Given a matrix 'a' of dimension $n \times m$ and 2 coordinates $(l1, r1)$ and $(l2, r2)$. Return the sum of the rectangle from $(l1, r1)$ to $(l2, r2)$.

print
 $m = 5 \rightarrow \text{rows}$
 $n = 4 \rightarrow \text{cols}$

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16
4	17	18	19	20

a

$(l1, r1) = (0, 1)$
 $(l2, r2) = (3, 2)$

Ques : Write a program to Print the transpose of the matrix entered by the user.

Done



Column-wise printing



Follow-up → 2D array →

1	2	3
4	5	6
7	8	9

Array → collection of variables



int, char, float

```
int arr[5];
```

```
char arr[4] =
```

Strings



char arrays

```
arr = {'r', 'a', 'g', 'h', 'a', 'v'}
```

What are strings?

↓
char arrays (mostly)

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Printing character array with loop

↓

This can be done, but is it not hectic?

↓

It is

→ Raghav is a teacher at PW.

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Now, let's print character array with null character

↓
done, it doesn't show in output

```
→ char arr = {'r', 'a', 'g', 'h', 'a', 'v', '\0'};
```

↓
raghav

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Initializing a string

Just how we make arrays! (Already done)

```
char str[] = {'P', 'h', 'y', 's', 'i', 'c', 's', ' ', 'W', 'a', 'l', 'l', 'a', 'h', '\0'};
```

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Initializing a string

A Better Way

```
char str[] = "PhysicsWallah";
```

What is the size of str? → ?

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Initializing a string

Assigning string literal with size:

```
char str[50] = "PhysicsWallah";
```

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Think...

Is the following code snippet correct?

```
char str[13] = "PhysicsWallah";
```

wrong, give str a min size of 14

arr → str

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Accessing individual characters

^{0 1 2 3 4 5 6 7 8 9 10 11 12}
 char str[] = "Physics Wallah";

Predict the output?

(a) printf("%c", str[5]); → c

(b) printf("%d", str[9]); → ascii value of l

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Modifying individual characters

```
char str[] = "Physics Wallah";
```

Predict str after:

(a) `str[0] = 'M';` *M hysics Wallah*

(b) `str[1] = 97;` *Maysics Wallah*

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Input and Output of string without loop

(puts & gets)

↓
using '%s' format specifier

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Can we use string (character arrays) as pointers?

```
char str[] = "PhysicsWallah";  
char *ptr = str;
```

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We get another way of initialising strings

```
char *ptr = "Physics Wallah";
```

Note: Such direct initialisation using pointers results in a read-only memory allocation of character arrays and hence, causes **undefined behaviour** when we try to change the characters.

```
ptr[0] = 'm';     Error!
```

Interesting thing about such initialization

```
char str[] = "Physics Wallah";  
printf("%s", str);  
str = "College Wallah"; Error!
```

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Interesting thing about such initialization

```
✓ char *ptr = "Physics Wallah";  
  printf("%s \n", ptr);  
✓ ptr = "College Wallah"; Works perfectly!  
  printf("%s", ptr);
```

ptr → "Physics Wallah"
 ↘ "College Wallah"

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Why does this happen?

Pointers change the address to which they point after initialising a new character array!

```
char *ptr = "Physics Wallah";  
printf("Address 1: %p \n", ptr);  
ptr = "College Wallah";  
printf("Address 2: %p", ptr);
```

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Using heap memory

Storing str in writable heap segment

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Implement: Copy one string to another

```
char s1[] = "Physics Wallah";  
char *s2 = s1;
```

```
// Let's change in s1.  
s1[0] = 'M';  
printf("%s", s2);
```

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Making our tasks easy!

Useful functions for C already in standard library!

strlen(char *str)	Returns the length of string
strcpy(char *s1, char *s2)	Copies the contents of string s2 to string s1
strcat(char *s1, char *s2)	Concat s1 string with s2 and stores the result in s1
strcmp(char *s1, char *s2)	Compares the two strings
strncpy(char *s2, char *s1, int len)	Copy substring of size len starting from s1 character pointer into s2.

H.W. Reverse a given string

↓

char str[] = "raghav";

↓

n = 7

↓

i=0, j=6

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Structures

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What are Structures?

↓
User defined data type

```
struct student{  
    int rno;  
    char grade;  
    float percent;  
};
```

da bbe int, float, char



Student

Every student has some common properties

↓
rno, percent, grade

```
int x;  
x = 5
```



```
student x;
```



int
↓
data type
blueprint / class

```
int x;
```



```
struct student{
    int rno;
    float percent;
    char grade;
};
```

```
✓ struct student s; // declaration
✓ s.grade = 'A';
✓ s.rno = 88;
✓ s.percent = 95.4;
return 0;
```

grade	rno	percent
A	88	95.4

s

Why Structures?

2D array

↓

Raghav	90.1	B	76
Urvi	95.4	A	88
Harsh	93.2	B	82
Sanket	96.8	A	13
	float per	grade	rnno

f	f	f
i	i	i
c	c	c

Array approach vs Structures



Same data type ke
multiple objects ko
store karna



Diff. data types for
a specific class

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Ques : What should be preferred to store 10 floats in a memory? ~~Array~~ or structure?

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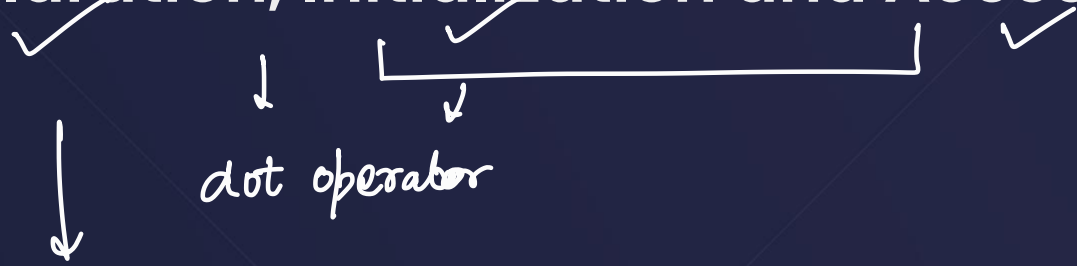
Ques : State true or false ✓

An array should be used to store **dissimilar** elements, and a structure to store similar elements.

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Structure variables

Declaration, Initialization and Accessing



Struct student s1;

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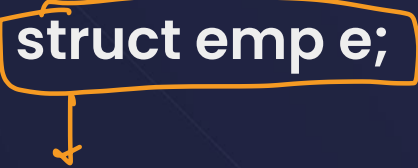
Ques : Create a structure type 'book' with **name, price** and **number of pages** as its attributes → *Classwork*

```
struct book{
    char name[20];
    float price;
    int noOfPages;
};
```

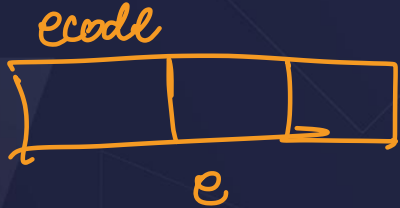
```
struct book b1;
b1.price = 487.5
b1.noOfPages = 500;
strcpy (b1.name, "Secret Seven");
```

Ques : Find the error

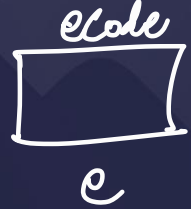
```
struct emp{
    int ecode;
    struct emp e;
};
```



struct emp e;



```
struct emp{
    int ecode;
};
struct emp e;
```



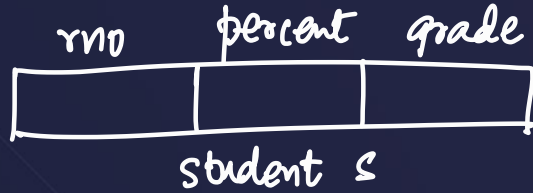
Ques : Create a structure type 'Person' with name, salary and age as its attributes. Declare and initialize 2 variables for this. Print the name of first person and age of the other.

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How are structure elements stored?

↓

Memory Allocation



↓
10 bytes

Contiguous

int x; → 4 bytes

float f; → 4 bytes

char ch → 2 bytes

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Typedef in struct



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Array of Structures

Why?

↓
50 students ka data, → height, mo, grade

↓
`int arr[50];`

`student arr[50];`

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Array of Structures

Declaration and Access



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Features of structures

Copying one structure variable to another of the same type



Student s2 = s1

int x = 4; → 4
x

int y = x;

↓
6
4
y = 6
y

Features of structures

Nesting one structure within another structure

```

struct car {
    char type [20];
    int price;
    float topspeed;
};
    
```

Supercar
 ↓
 car ✓ + extra
 ↓
 chassis & spoilers

Macbook air → touch id, haptic trackpad, airdrop

Macbook Pro → air ✓ + HDMI + SD card

Ques :

`maruti.engine.bolts = 25 ;`



Which of the following is True?

1. structure bolts is nested within structure engine *True*
2. structure engine is nested within structure maruti *True*
3. structure maruti is nested within structure engine *F*
4. structure maruti is nested within structure bolts *F*

legendary Po

Po

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Features of structures → passed by value

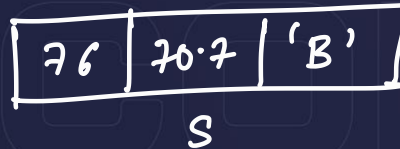
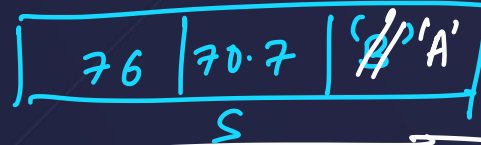
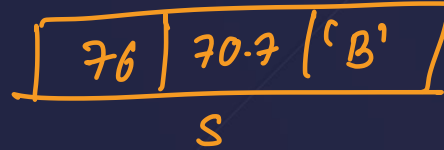
A structure variable can be passed to a function

```

void print(student s){
    printf("%d\n",s.rno);
    printf("%f\n",s.percent);
    printf("%c\n",s.grade);
}

void change(student s){
    s.grade = 'A';
}

int main(){
    ✓ student s = {76,70.7,'B'};
    ✓ print(s);
    ✓ printf("%c\n",s.grade);
    ✓ change(s);
    ✓ printf("%c\n",s.grade);
    return 0;
}
    
```



76
70.7
B
B
B

Features of structures

Structure pointers



for pass by reference

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What are Unions? → Structures only

↑ space efficient

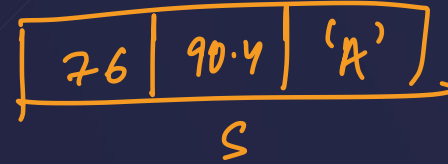
```
union student{
    int rno;
    char grade;
    float percent;
};
```



struct → 10 bytes
union → 4 bytes



but with a discipline



At a time only one
attribute can be
accessed → in fact stored

Finish :



Wait,

File Handling

DMA

Preprocessor, Macros



Youtube



College Wallah



Playlist



C Programming Course



File Handling

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Maza aa gaya → Lecture Complete

THANK YOU

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